significant gaps remain between what we need to measure and what can be practically automated. Cost, technology limitations, and practical constraints often require combining automated sensing with manual data collection and third-party data sources to create a complete picture of your physical system. In my home digital twin, fully automating every measurement would cost thousands of dollars and require extensive modifications to existing infrastructure. Industrial digital twins face similar trade-offs at larger scales: a chemical plant might have thousands of manual gauge readings collected during operator rounds, while a smart city might rely on citizen-reported issues to supplement its sensor network. The key to success is to strategically combine different data collection methods based on the value of the information, the cost of automation, and the required update frequency. High-value, rapidly changing parameters justify expensive automated sensing, whereas stable parameters that change slowly can rely on periodic manual collection. Environmental conditions affecting your system may be better sourced from external providers than duplicating measurement infrastructure. By thoughtfully mixing these approaches, you can build a digital twin that delivers insights without breaking budgets or requiring impractical deployments.

3.4.1 Edge processing

Rather than transmitting raw sensor data to remote servers for analysis, edge devices perform real-time processing locally, extracting meaningful insights and transmitting only relevant information. This approach dramatically reduces bandwidth and power requirements and can enable autonomous decision-making even when network connectivity is intermittent or unavailable. My main water meter provides a perfect example of where edge processing is valuable in sensing the real world.

EXAMPLE: ANALOG METER READING

I can convert my water meter from analog to digital by taking a photograph of it and using OCR to extract the current reading. The challenge, shared in many real world scenarios, is that the use of low power, low bandwidth, long range battery powered sensors means that transmitting every image to the cloud for processing is often not feasible.